

Akshitha Shetty

ROS 2 Developer — Autonomous Robotics Engineer — UAV Systems

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Professional Summary

Final-year Robotics and Automation Engineering student specializing in ROS 2 autonomous systems, UAV development, and hardware-software integration. Experienced in SLAM, path planning, computer vision, and mechanical design, with hands-on experience building real-world robotic and UAV systems for national-level competitions including ISRO BAH 2026 and MAHE Hackathon 2026.

Technical Skills

Robotics & Autonomy	ROS 2 (Humble), Nav2, SLAM Toolbox, Gazebo, RViz, URDF
Programming	Python, C++, C, Bash
CAD & Simulation	SOLIDWORKS, MATLAB/Simulink, Fusion 360
Computer Vision	OpenCV, YOLOv8, ArUco Marker Detection
UAV Systems	Pixhawk, ArduPilot, MAVLink, Mission Planner
Tools & Platforms	Git, Linux, Raspberry Pi, Arduino

Education

Sahyadri College of Engineering and Management, Mangaluru	2023–2027
B.E. Robotics and Automation Engineering	CGPA: 7.01

Technical Projects

ICE-PATH: Lunar Ice Detection & Rover Traverse Planning *ISRO BAH 2026*

- Fused L/S-band DFSAR radar, terrain, and illumination models into an explainable “Ice Confidence Index” for the lunar south pole, combining multi-source geospatial data for high-reliability mapping.
- Implemented weighted A* rover traverse planning with dynamic Safety-First and Shortest-Path profiles; demonstrated traversal feasibility across permanently shadowed regions.

Intelligent Autonomous Warehouse Picking Robot *MAHE Hackathon 2026*

- Developed an autonomous warehouse robot using ROS 2, Nav2, OpenCV, and ArUco marker detection as part of Team SudoSquad, competing at national level.
- Built a multi-frame marker filtering pipeline and a cost-based mission planner enabling dynamic re-planning and task queue updates in under 200 ms.

Hybrid Autonomous Navigation System *Sahyadri College of Eng. & Mgmt.*

- Developed a hybrid ROS 2 pipeline bridging Nav2 global planning with local vision-based guidance for AGVs; implemented HSV line following, custom Action Servers, and Twist multiplexing.
- Simulated a URDF robot model in Gazebo, tuning costmap parameters and recovery behaviors for robust indoor navigation.

AgriDrone Audit: UAV-Based Crop ET Sensing *Smart India Hackathon 2024*

- Developed a UAV-based evapotranspiration sensing and irrigation water accounting system using YOLOv8 and OpenCV, achieving ~85% crop detection accuracy on aerial imagery.
- Designed a scalable processing pipeline to classify crop health zones and flag irrigation anomalies from drone footage.

Experience

Project Intern — NIT Karnataka *Feb 2025–Mar 2025*

- Designed a SOLIDWORKS catapult launch and parachute recovery mechanism for a 1.5 kg fixed-wing UAV; performed CG balancing and flight-dynamics modeling in MATLAB/Simulink.
- Integrated autonomous mission execution with real-time GCS telemetry using Mission Planner and ArduPilot.

Core Team Member & Documentation Lead — Team Challengers *Jun 2024–Present*

- Designed, built, and piloted aeromodels and UAVs for national-level events including IIT Kanpur Techfest.
- Led technical documentation for Aerophilia 2025 and conducted hands-on STEM aeromodeling workshops for junior students.

Certifications

- **Python Foundation** — SpringBoard **MATLAB Onramp** — MathWorks **Simulink Onramp** — MathWorks